

Session Objectives

**NEW YORK INSTITUTE
OF TECHNOLOGY**

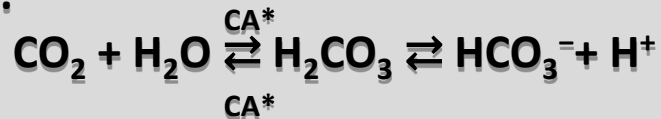
College of Osteopathic
Medicine

At the conclusion of the session, the student should be able to:

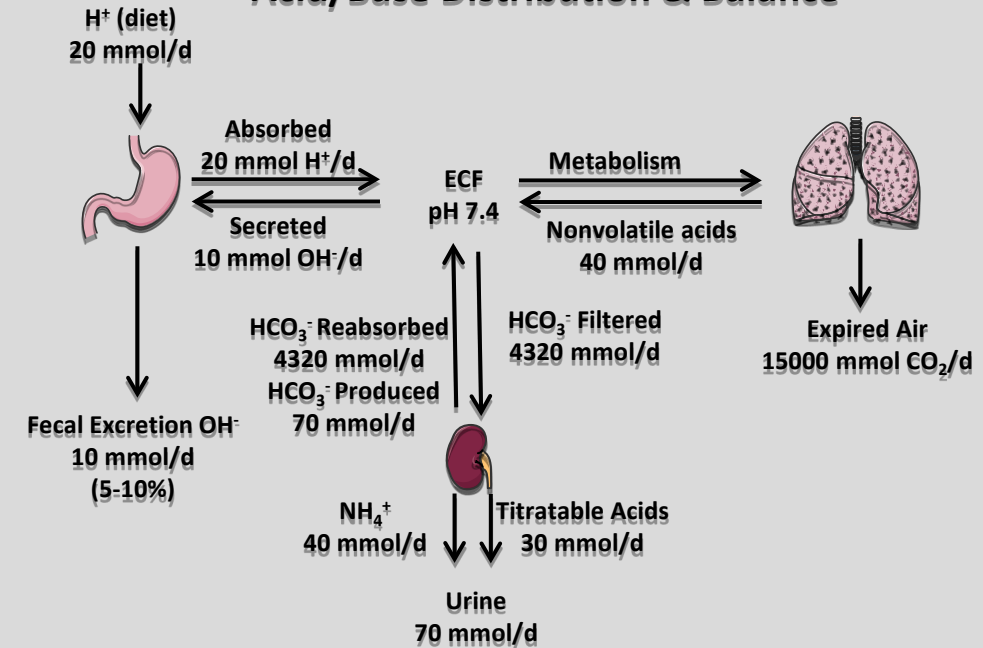
- 1. Define & describe the major sources of acids and bases in the body, including metabolic processes**
- 2. Explain the fundamentals of acid/base balance in the body.**
- 3. Define & describe the relative roles of the various segments of the tubule in the renal processing of H^+ and HCO_3^- .**
- 4. Define & describe the major regulatory sites and mechanisms of renal processing of H^+ and HCO_3^- .**

Acid-Base Balance and the Renal Handling of Acid

- Acid-base imbalances are among the most important problems confronting clinicians.
- Kidneys & lungs are responsible for regulating acid-base balance.
- Via independent control of the major components of the body's buffering system: CO_2 & HCO_3^- .
- Lungs excrete the largest amount of CO_2 formed by metabolism
- Kidneys are crucial for excreting *nonvolatile* acids.
- CO_2 production: Largest source of acid
- Occurs during oxidation of carbohydrates, fats & and most amino acids
- Lungs excrete a large amount of CO_2 by diffusion across the alveolar-capillary barrier.



- Kidneys monitor the acid-base parameters of the ECF and adjust acid secretion to maintain the pH of ECF (7.37–7.42).
- Kidneys play an essential role in acid-base equilibrium by neutralizing *nonvolatile* acids.
- Characteristic symptoms of renal failure:
 - Severe acidosis caused by acid retention



Volatile acids: Acids derived directly from the hydration of CO_2 .
Nonvolatile/Titratable acids: Acids NOT derived directly from the hydration of CO_2 (Ex: H_2PO_4^-).

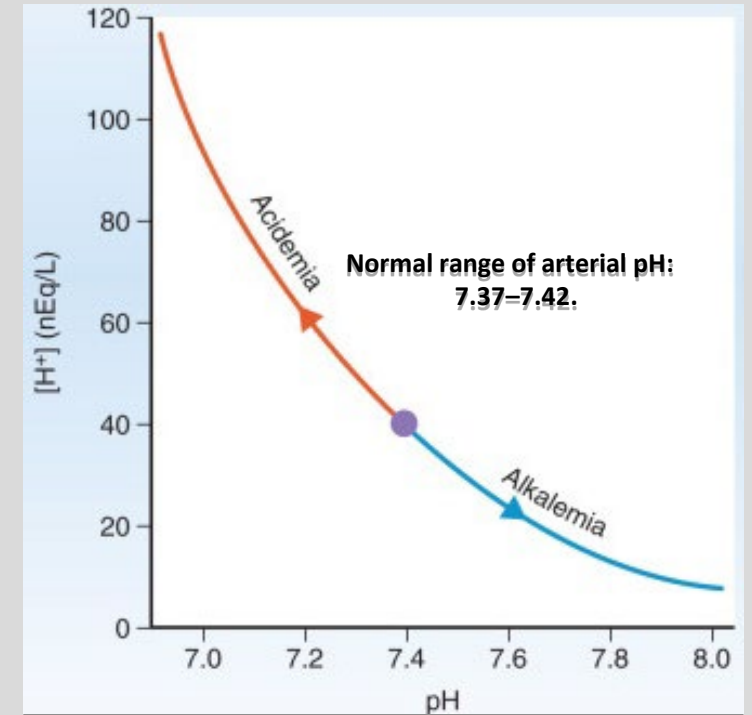
$\text{CO}_2/\text{HCO}_3^-$: Major buffering system in body.

$\text{NH}_3/\text{NH}_4^+$: Major buffer system SYNTHESIZED in the kidney.

*CA: Carbonic anhydrase

Kidneys Secrete Generated-Nonvolatile Acids

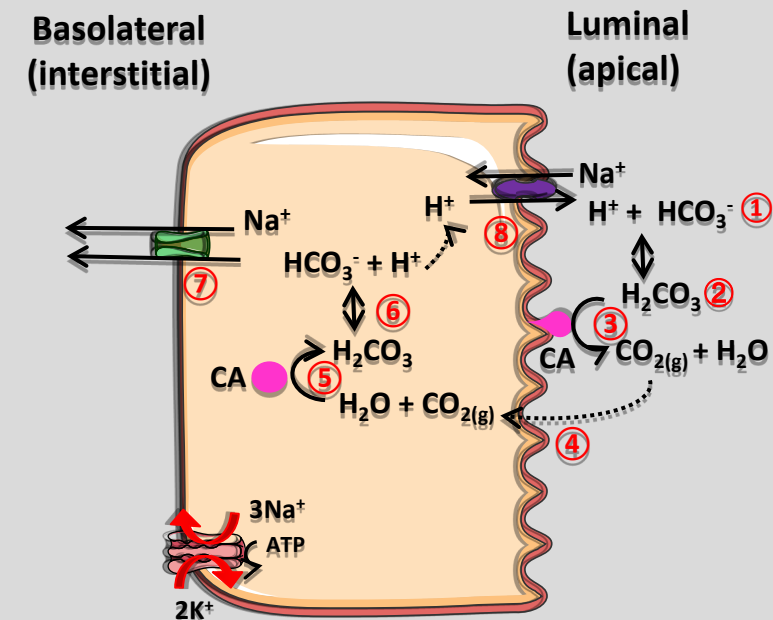
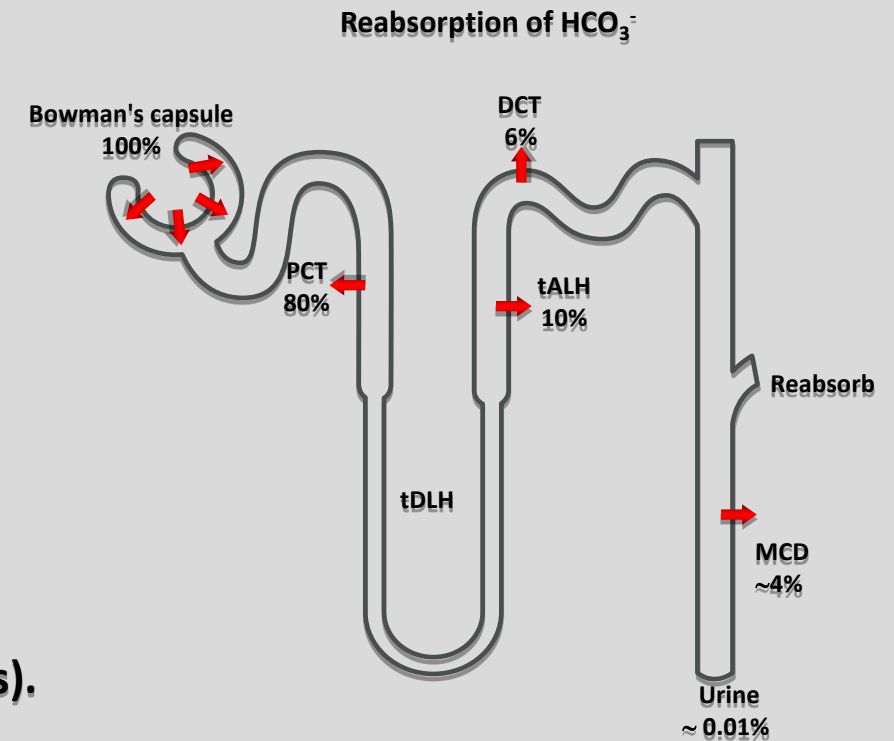
- The lungs excrete large amounts of a potential acid in the form of CO_2 .
- Kidneys play an essential role in normal acid-base equilibrium.
- They are the sole effective route for neutralizing nonvolatile acid via secretion of acid into the urine.
- Prior to renal neutralization, it must retrieve ALL of filtered HCO_3^- .
- Daily filtered load of $\text{HCO}_3^- = 4320 \text{ mmol}$.
- If this HCO_3^- was left behind in the urine $\rightarrow$ blood acid load of 4320 mmol
- Result: Catastrophic metabolic acidosis
- 70 mmol/d acid load: Produced by metabolism, diet, and intestinal losses.
- Would correspond to a urinary pH ~ 1.3 if not regulated.
- The lowest urinary pH is ~ 4.4
- This corresponds to an $[\text{H}^+]$ three orders of magnitude lower than required to excrete the 70 mmol/day of nonvolatile acids.
- The kidneys solve this problem by binding H^+ to:
 - Nonvolatile Acids
 - $\text{NH}_3/\text{NH}_4^+$ (NH_3 (renal origin) + $\text{H}^+ \rightleftharpoons \text{NH}_4^+$)
 - HCO_3^-



$$\text{Net } [\text{H}^+]_{\text{U excretion}} = [\text{H}^+]_{\text{Ex bound to Nonvolatile acids}} + [\text{H}^+]_{\text{Ex bound to } \text{NH}_3} - [\text{HCO}_3^-]_{\text{Ex}}$$

Renal Control of HCO_3^-

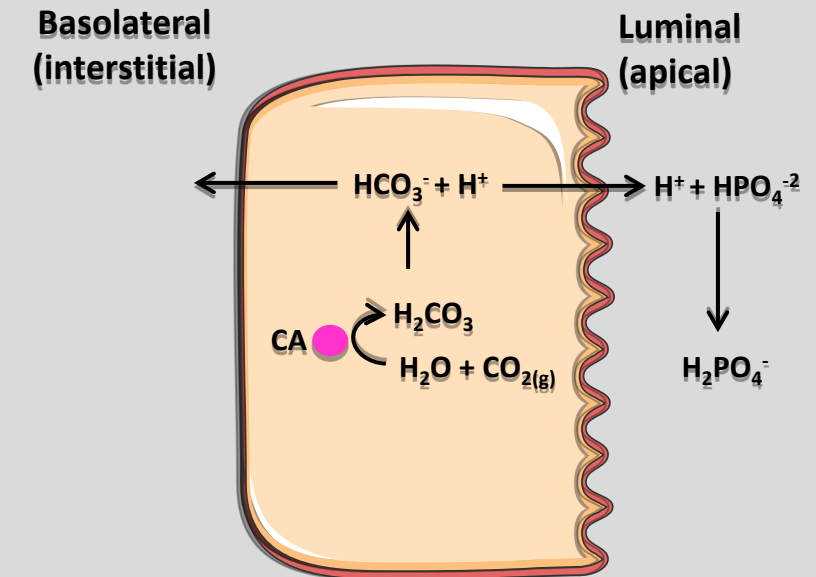
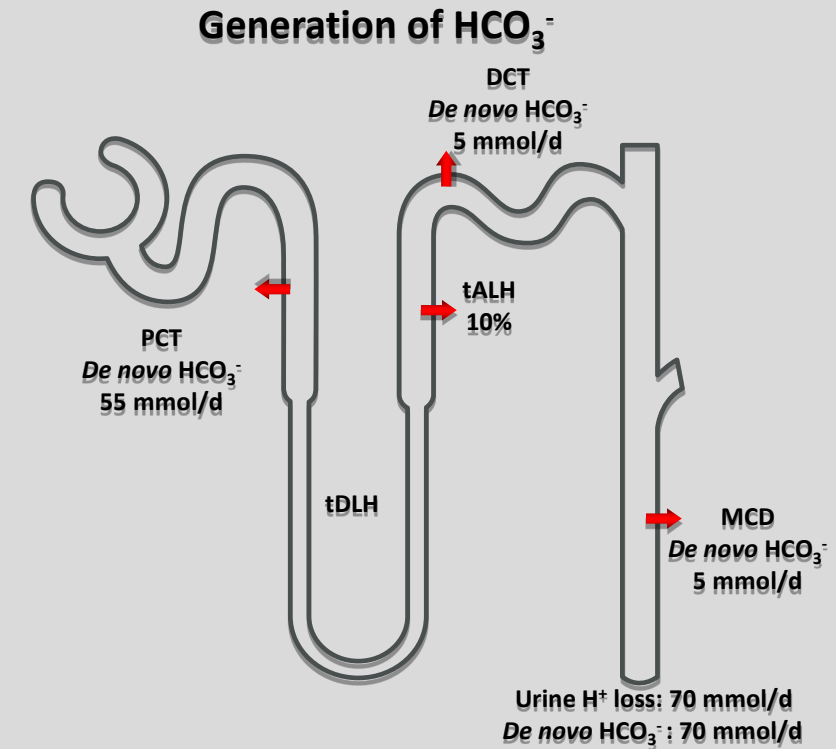
- HCO_3^- reabsorbed at specialized sites along the nephron.
- Basic mechanism of HCO_3^- reabsorption is the same:
 - H^+ is transported into the lumen by the tubule cell ①.
 - It is titrated w/ HCO_3^- to form H_2CO_3 (carbonic acid) ②.
 - H_2CO_3 dissociates to yield H_2O and $\text{CO}_{2(g)}$ ③.
 - Rx ($\text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_{2(g)}$) is too slow to convert the FL of HCO_3^- .
 - Catalyzed by carbonic anhydrase (CA).
- CA:
 - Nearly ubiquitous family of Zn-containing enzymes (at least 16 members).
 - Catalyzes the BIDIRECTIONAL interconversion of CO_2 and HCO_3^- .
 - CA I: Mainly in the cytoplasm of RBC.
 - CA II: Ubiquitous cytoplasmic enzyme ⑤.
 - CA IV: Membrane-linked enzyme found on the outer surface of the apical membrane of the renal proximal tubule ③.
- $\text{CO}_{2(g)}$ will diffuse across the apical membrane ④ and get converted to H_2CO_3 ⑤
- H_2CO_3 dissociates to HCO_3^- & H^+ ⑥.
- HCO_3^- is transported to circulation by a $\text{Na}^+/\text{HCO}_3^-$ cotransporter ⑦ & H^+ is secreted into luminal space via a $\text{Na}^+/\text{antiporter}$ ⑧
- A hallmark of many CAs is their inhibition by sulfonamides (e.g., acetazolamide).



Titration of Filtered, Non-HCO₃⁻ Buffers

- Secreted H⁺ interacts with buffers other than HCO₃⁻ & NH₃ (titration of non-NH₃, non-HCO₃⁻ buffers).
- React with HPO₄⁻², creatinine & urate to their conjugate weak acids (HB)

$$\text{H}^+ + \text{B}^- \rightarrow \text{HB} \text{ (titratable acid)}$$
- Kidneys excrete H⁺-bound phosphate in the urine.
- ↓ pH inhibits apical Na/phosphate cotransporters (NaPi) in the PT
- ∴ transports H₂PO₄⁻ less effectively than HPO₄⁻
- Each H⁺ transferred to the lumen to titrate HPO₄⁻² → generation of new HCO₃⁻
- HCO₃⁻ is subsequently transferred to the blood.
- Contribution to net acid excretion is dependent on 3 factors:
 - Amount of the buffer in the filtrate and urine (e.g., FL of HPO₄⁻²).
 - pK of the buffer:
 - To be most effective at accepting H⁺, buffer should have a pK that is between the pH of the glomerular filtrate and the pH of urine. (e.g., phosphate, creatinine, urate)
 - pH of the urine:
 - The lower the urinary pH, the more protonated the buffer and the greater the amount of acid excreted with this buffer.



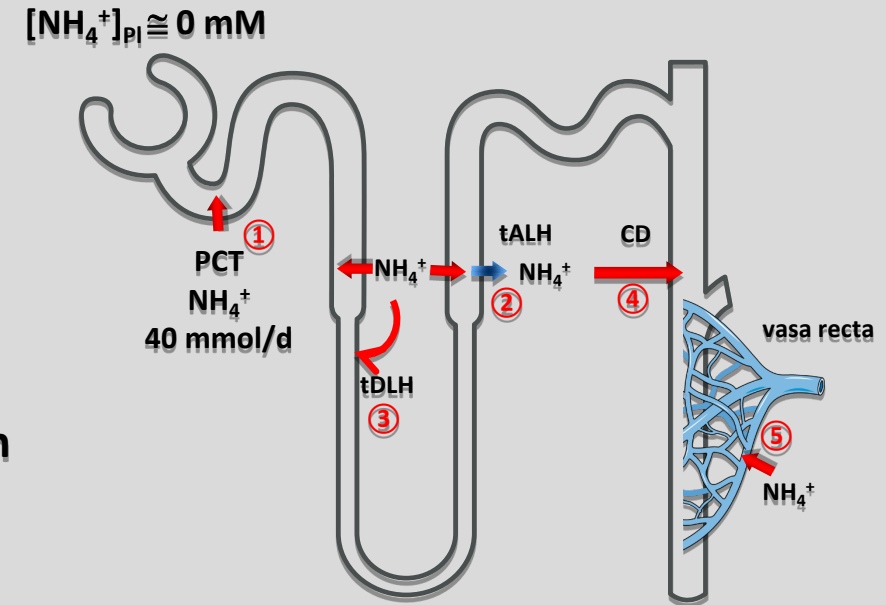
Titration of Filtered and Secreted NH_3

- NH_3 : Third class of acceptors of luminal H^+ .
- Unlike HCO_3^- or HPO_4^{2-} , filtration contributes only a negligible quantity of NH_3 .
- $[\text{NH}_3]_{\text{PI}}$ concentration is exceedingly low.
- $\text{NH}_3/\text{NH}_4^+$ excretion is the result of five processes:
 - The PCT secretes $\sim 40 \text{ mmol/day}$ of NH_4^+ ①
 - tALH reabsorbs NH_4^+ & deposits it in the interstitium ②
 - Some of this interstitial NH_4^+ recycles back to the proximal tubule and thin descending limb (tDLH) ③
 - Interstitial NH_4^+ enters the lumen of the collecting duct ④
 - Interstitial NH_4^+ enters the vasa recta and leaves the kidney ⑤

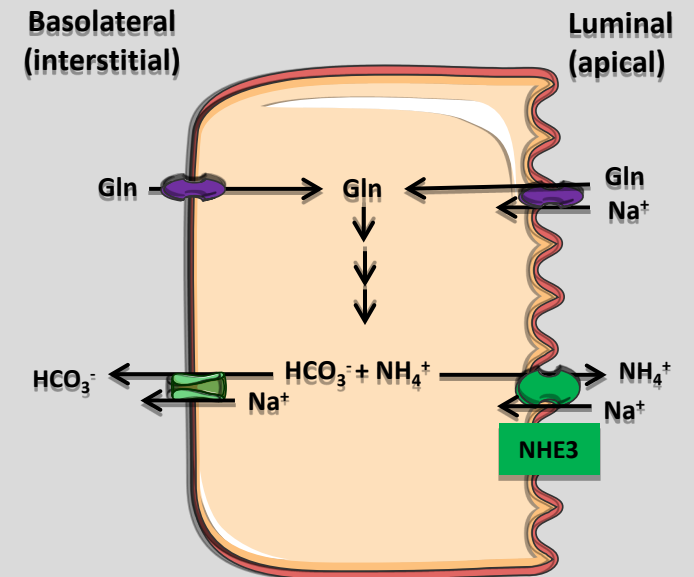
PCT:

- Urinary NH_3 is derived from diffusion into the lumen from the PCT ①.
- Conversion of glutamine to α -ketoglutarate (α -KG) generates two NH_4^+ ions, which form two 2NH_3 & 2H^+ .
- NH_3 enters the lumen via the apical Na^+/H^+ exchanger (NHE3).
- In addition, the metabolism of α -KG will also generate a HCO_3^- ion.
- This “new” HCO_3^- enters the blood.

Excretion of Ammonium (NH_3)



Proximal Tubules



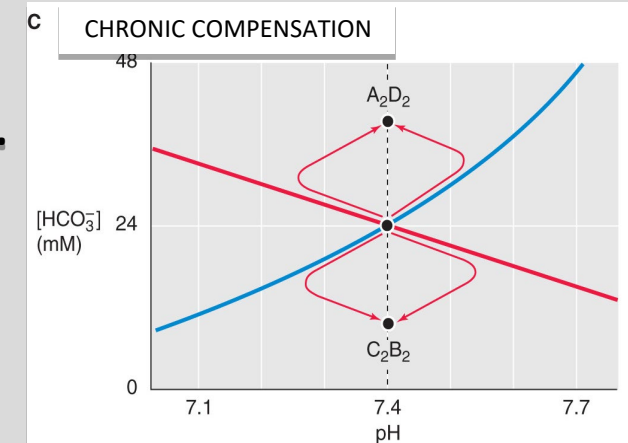
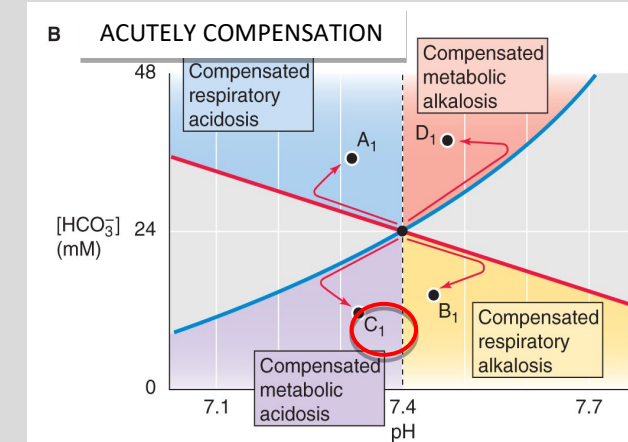
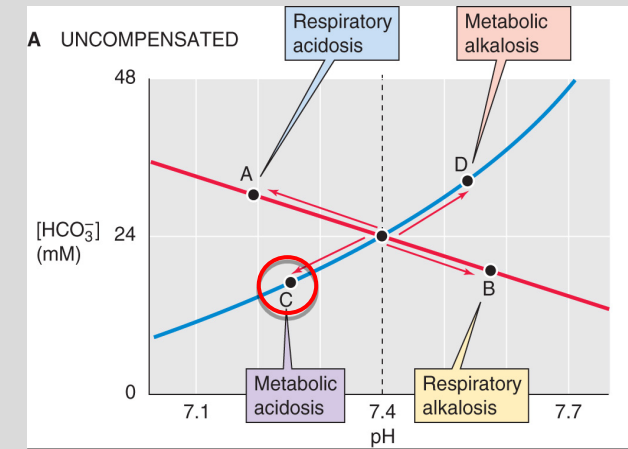
Regulation of Renal Acid Secretion

Respiratory acidosis:

- Stimulates H^+ secretion in three ways:
 - Proximal Tubules:** Acute respiratory acidosis ($\uparrow P_{CO_2} \rightarrow$ insertion of H^+ pumps into apical membranes $\rightarrow \uparrow H^+$ secretion
 - Distal Tubule:** Acute respiratory acidosis $\rightarrow$ insertion of H^+ pumps into the apical membranes $\rightarrow \uparrow H^+$ secretion.
 - Chronic respiratory acidosis:** Upregulation of activities of apical NHE3 and basolateral NBCe1 in proximal tubule.

Metabolic acidosis:

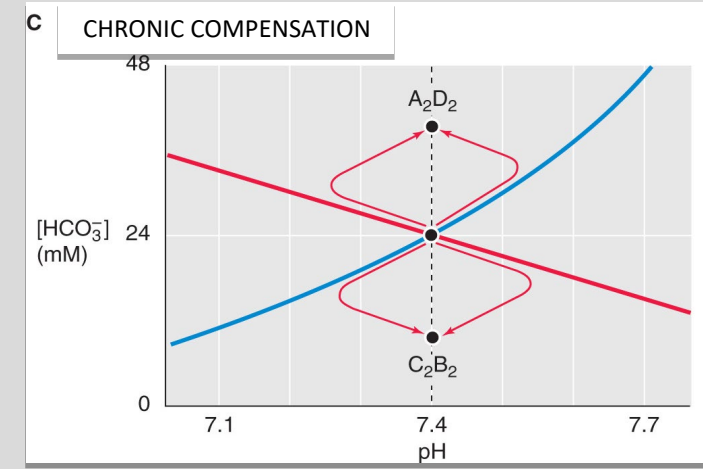
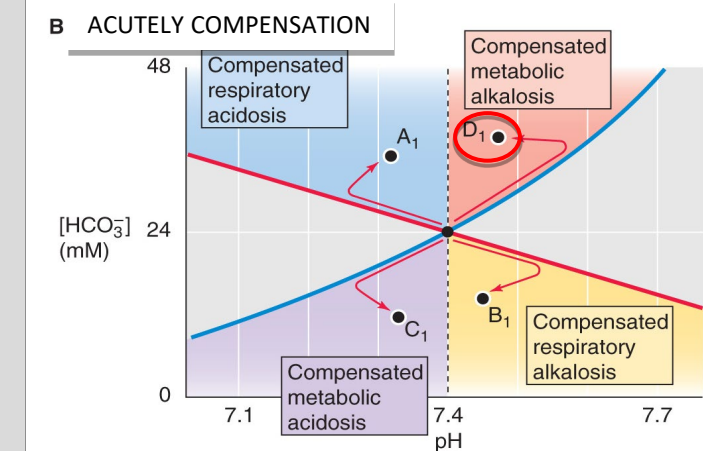
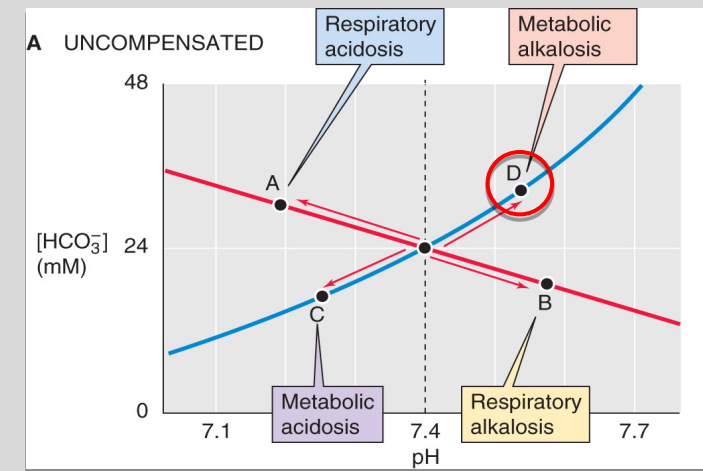
- Acute metabolic acidosis:**
 - Stimulates both H^+ secretion and NH_3 production
 - Pulmonary compensation:** Increased alveolar ventilation $\rightarrow CO_2$ expiration $\rightarrow$ corrected $[HCO_3^-]/[CO_2]$ ratio
 - Renal compensation:**
 - Proximal Tubule:** $\downarrow [HCO_3^-] \rightarrow \uparrow H^+$ secretion
 - Distal Tubule (Intercalated cells):** $\uparrow H^+$ pump insertion into apical membranes.
- Chronic metabolic acidosis:**
 - Proximal tubule:** Responses are similar to chronic respiratory acidosis.
 - $\uparrow Na^+$ -citrate uptake $\rightarrow \uparrow$ Citrate metabolism $\rightarrow \uparrow HCO_3^-$ production
 - $\downarrow [Citrate]_{PI} \rightarrow \downarrow [Ca^{++}\text{-Citrate}]_{EX} \rightarrow \uparrow [Ca^{++}]_U \rightarrow$ Renal calcium stones
 - $\uparrow$ increase NH_4^+ excretion.



Regulation of Renal Acid Secretion

Metabolic Alkalosis

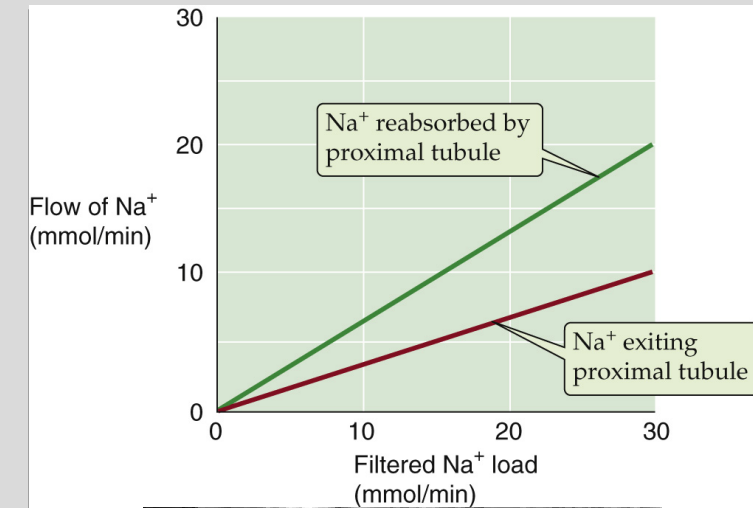
- Proximal Tubule:
 - $\uparrow[\text{HCO}_3^-]_{\text{pl}} \rightarrow \downarrow[\text{H}^+]_{\text{pl}} \rightarrow \downarrow \text{H}^+ \text{ efflux in the PT}$
- Collecting Duct (ICT & CCT):
 - α -intercalated cells:
 - Switch from luminal $[\text{H}^+]$ secretion to $[\text{HCO}_3^-]$ secretion.
 - Via apical H^+ pump and basolateral $\text{Cl}^-/\text{HCO}_3^-$ exchanger
 - β -intercalated cells:
 - Increased proportion of β IC from α IC.
 - They have the opposite apical/basolateral distribution of H^+ pumps and $\text{Cl}^-/\text{HCO}_3^-$ exchangers $\rightarrow \uparrow[\text{HCO}_3^-]_{\text{Lumen}}$



Regulation of Renal Acid Secretion

Glomerular Filtration Rate:

- $\uparrow$ GFR increases HCO_3^- delivery to the tubules $\rightarrow$ enhances reabsorption (glomerulotubular balance).
- Similar to Na^+ balance
- $\uparrow \text{HCO}_3^-$ delivery $\rightarrow \uparrow \text{pH}_{\text{Lumen}} \rightarrow \uparrow \text{NHE3 \& H pumps in PT} \rightarrow$ enhanced H^+ secretion
- $\uparrow$ Flow also increases the shear force that acts on the central cilium present on every proximal-tubule cell.
- The more the cilium bends $\rightarrow$ the greater the signal to increase the reabsorption of solutes.
- GT balance is important because it minimizes HCO_3^- loss
- $\therefore \uparrow \text{GFR} \rightarrow$ Development of a metabolic acidosis.
- Conversely, GT balance of HCO_3^- reabsorption also prevents metabolic alkalosis when GFR decreases.
- Flow dependence of HCO_3^- reabsorption also accounts for the stimulation of H^+ transport that occurs after uninephrectomy (i.e., surgical removal of one kidney)
- GFR in the remnant kidney rises in response to the loss of renal tissue.



Primary cilium: Flow sensor

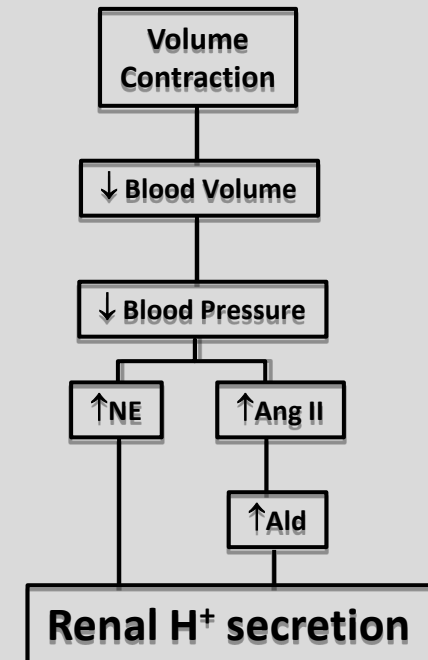
Regulation of Renal Acid Secretion

Volume Contraction:

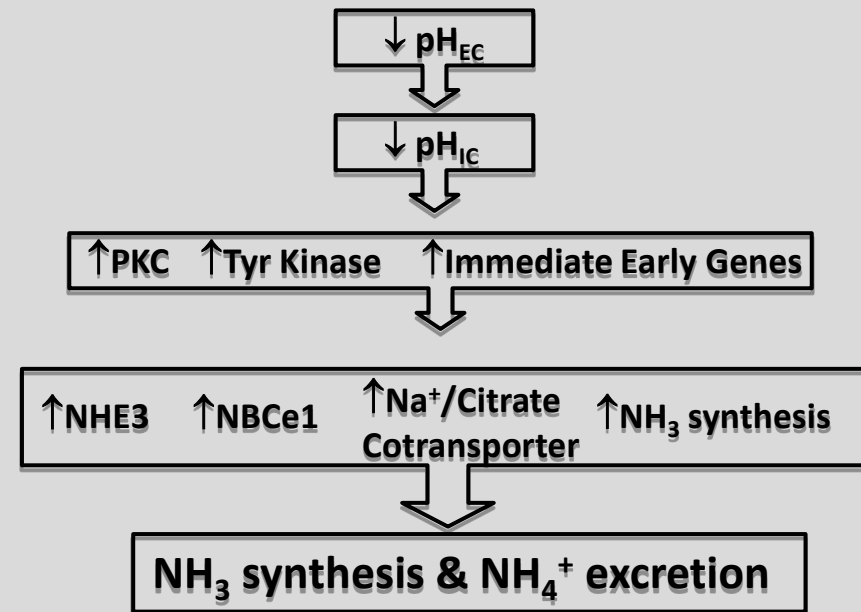
- $\downarrow$ Extracellular volume $\rightarrow \uparrow$ RAAS & SNS activity $\rightarrow$ renal H^+ secretion
- Proximal Tubule:
 - Ang II and NE stimulate Na^+/H^+ exchanger in the proximal tubule.
 - Volume contraction $\rightarrow Na^+$ reabsorption & H^+ secretion.
- Distal Tubule:
 - Ang II stimulates acid secretion by α -intercalated cells.
- Chronic volume depletion increases aldosterone levels
 - Collecting Ducts: Aldo enhances H^+ secretion in cortical and medullary collecting ducts.
 - Associated with changes in plasma K^+

Volume Expansion:

- $\uparrow$ Extracellular volume $\rightarrow \downarrow$ RAAS & SNS activity $\rightarrow$ renal H^+ retention
- Thus, the regulation of effective circulating volume takes precedence over the regulation of plasma pH.



Chronic Acidosis in Proximal Tubule



Regulation of Renal Acid Secretion

Hypokalemia:

- K⁺ depletion → ↑renal H⁺ secretion → metabolic alkalosis.

Proximal Tubule:

- Chronic acidification:
 - ↑Na⁺/H⁺ exchange (apical) & Na⁺/HCO₃⁻ (basolateral) cotransport → ↓urinary pH
 - K⁺ depletion → NH₃ synthesis & NH₄⁺ excretion

Collecting Duct (ICT/CCT):

- ↑apical K⁺/H⁺ exchanger (αIC) → ↑H⁺ secretion

Hyperkalemia

- Often associated with metabolic acidosis.

Proximal Tubule:

- ↑pH_{IC} → lower NH₄⁺ synthesis → reduced NH₄⁺ excretion

Thick Ascending Limb of the loop of Henle:

- [K⁺]_L → competition w/ NH₄⁺ for uptake by NKCC & ROMK → ↓NH₄⁺ reabsorption → ↓NH₄⁺ in medullary interstitium → ↓NH₃ for diffusion into collecting duct

Medullary collecting duct.

- ↑[K⁺ +] in the medullary interstitium → competes with NH₄⁺ uptake → ↓NH₄⁺ excretion → Acidosis

Regulation of Renal Acid Secretion

Corticosteroids:

- Glucocorticoids (GC) and mineralocorticoids (MC) stimulate acid secretion.
- Chronic adrenal insufficiency → acid retention → metabolic acidosis (life-threatening)
- Both GCs & MCs H^+ secretion at different sites along the nephron.

Glucocorticoids (e.g., cortisol):

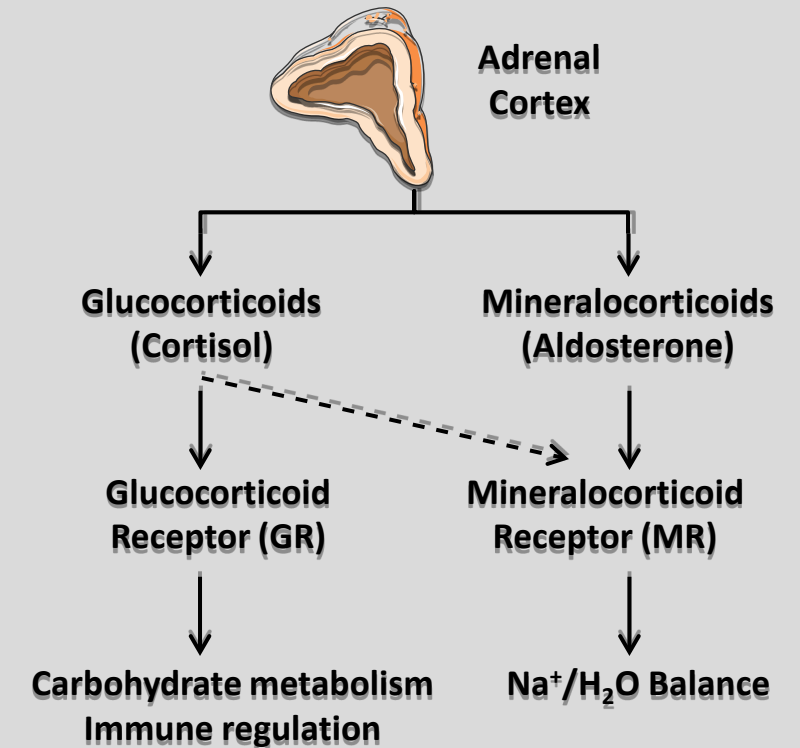
- $\uparrow Na^+/H^+$ exchanger (proximal tubule) → $\uparrow H^+$ secretion
- $\downarrow PO_4^{3-}$ reabsorption → $\uparrow$ luminal availability of buffer anions for titration

Mineralocorticoids (e.g., aldosterone):

- $\uparrow H^+$ secretion by:
 - Directly $\uparrow$ secretion (CD) via $\uparrow$ activity of the H^+ pump (apical) & basolateral Cl^-/HCO_3^- exchanger.
 - Indirectly $\uparrow H^+$ secretion by $\uparrow Na^+$ reabsorption in the CD → $\uparrow$ lumen-negative voltage → $\uparrow H^+$ pump in α -intercalated cells → Acid secretion
 - Chronic elevation & accompanied by high Na^+ intake → K^+ depletion → indirectly increase H^+ secretion

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



Regulation of Renal Acid Secretion

Diuretics

- The effects of diuretics on renal H^+ secretion vary among diuretics.
- Depends on site & mechanism of action.
- Diuretics fall broadly into two groups:
 - Promote excretion of alkaline urine
 - CAI and K^+ -sparing diuretics
 - CAI inhibitors $\rightarrow$ alkaline urine via $\downarrow H^+$ secretion.
 - Greatest effect is in the PT
 - $\downarrow H^+$ secretion by tALH & DCT
 - K^+ -sparing diuretics (amiloride & triamterene):
 - Inhibit ENaC in CD $\rightarrow$ Lumen is more positive $\rightarrow$ difficult for H^+ secretion into the lumen.
 - Spironolactone: $\downarrow H^+$ secretion by blocking Aldo.
 - Promote excretion of acidic urine (often inducing alkalosis)
 - Loop & Thiazide diuretics
 - They act by:
 - Volume contraction
 - Enhance Na^+ delivery to the CD $\rightarrow \uparrow Na^+$ uptake $\rightarrow \uparrow$ lumen-negative voltage $\rightarrow \uparrow H^+$ secretion
 - K^+ wasting $\rightarrow$ Hypokalemia $\rightarrow \uparrow H^+$ secretion

Nephron Site	Drug	Diuretic Class	Target
PCT	Acetazolamide	Carbonic Anhydrase Inhibitors (CAI)	Carbonic Anhydrase
tALH	Furosemide Bumetanide Ethacrynic acid	Loop diuretics	$Na^+/K^+/Cl^-$ cotransporter (NKCC2)
DCT	Metolazone	Thiazides	Na^+/Cl^- cotransporter (NCC)
CCT	Amiloride Triamterene	K^+ -Sparing	Na^+ channel (ENaC)
CCT	Spironolactone Eplerenone	K^+ -Sparing	Mineralocorticoid receptor (MR)
H_2O permeable segments	Mannitol	Osmotic diuretics	